Formula Sheet Part 3: Stationarity, AR Models, and ARCH

This is a two-column topic split from `All_Formulas_From_Chatnotes.md`.

Conventions

$$t, T, h, k, n, p, q, N$$

Variables: t is a time index; T is sample size or the final observed time; h is a forecast horizon; k is a lag or number of model parameters depending on context; n is a return horizon; p, q are lag orders; N is the number of assets.

$$F_t$$

Variables: F_t is the information set available at time t.

Returns may be written as decimals or percentages. Check the scale before applying VaR or standard deviation formulas.

Stationarity, Random Walks, ADF, and AR Models

Covariance stationarity

$$\begin{aligned}\mathbb{E}(Y_t) &= \mu \\ \mathrm{Var}(Y_t) &= \sigma^2 \\ \mathrm{Cov}(Y_t, Y_{t-j}) &= \gamma_j\end{aligned}$$

Variables: stationarity means constant mean, constant variance, and autocovariance depending only on lag j.

Stationarity rule of thumb

$$\begin{array}{rl}\text{log prices} & \text{often non-stationary} \\ \text{log returns} & \text{often stationary}\end{array}$$

Variables: this is a modelling guide, not a formal test.

Random walk

$$p_t = p_{t-1} + \epsilon_t$$

Variables: p_t is usually log price; ϵ_t is the shock.

Random walk successive substitution

$$p_t = p_0 + \epsilon_1 + \cdots + \epsilon_t$$

Variables: p_0 is the initial value.

Random walk moments

$$\mathbb{E}(p_t) = p_0, \qquad \mathrm{Var}(p_t) = \sigma^2 t$$

Variables: σ^2 is shock variance. If $p_0 = 0$, the note's simplified mean is $\mathbb{E}(p_t) = 0$.

Random walk with drift

$$p_t = \mu + p_{t-1} + \epsilon_t$$

Variables: μ is drift.

Random walk with drift successive substitution

$$p_t = p_0 + \mu t + \epsilon_1 + \cdots + \epsilon_t$$

Variables: drift accumulates linearly over time.

Random walk with drift moments

$$\mathbb{E}(p_t) = p_0 + \mu t, \qquad \mathrm{Var}(p_t) = \sigma^2 t$$

Variables: if $p_0 = 0$, the note's simplified mean is μt.

Unit-root AR form

$$y_t = \phi_1 y_{t-1} + \mathrm{error}_t$$

Variables: ϕ_1 is the autoregressive coefficient.

Unit root condition

$$\phi_1 = 1$$

Variables: a unit root implies random-walk-type non-stationarity.

ADF regression

$$\Delta y_t = c_t + \phi_c y_{t-1} + \beta_1 \Delta y_{t-1} + \cdots + \beta_p \Delta y_{t-p} + \epsilon_t$$

Variables: $\Delta y_t = y_t - y_{t-1}$; c_t is deterministic component; ϕ_c is the unit-root test coefficient; β_j controls lagged differences.

ADF coefficient relationship

$$\phi_c = \phi_1 - 1$$

Variables: $\phi_c = 0$ corresponds to $\phi_1 = 1$.

ADF hypotheses

$$\begin{array}{rl}H_0 : \phi_c = 0 & \text{unit root / non-stationary} \\ H_1 : \phi_c < 0 & \text{stationary}\end{array}$$

Variables: the ADF alternative is one-sided and negative.

ADF statistic

$$\mathrm{ADF} = \frac{\hat{\phi}_c}{\mathrm{se}(\hat{\phi}_c)}$$

Variables: $\hat{\phi}_c$ is the estimated ADF coefficient.

ADF decision rule

$$\mathrm{ADF} < \text{ADF critical value} \Rightarrow \text{reject } H_0$$

Variables: “less than” means more negative than the critical value.

Drift-only ADF deterministic component

$$c_t = c$$

Variables: c is a constant drift/intercept.

Trend ADF deterministic component

$$c_t = c_0 + c_1 t$$

Variables: c_0 is an intercept; c_1 is deterministic trend slope.

Martingale difference sequence

$$\mathbb{E}(\epsilon_t \mid F_{t-1}) = 0$$

Variables: ϵ_t has zero conditional mean given past information.

AR(1) model

$$y_t = c + \phi_1 y_{t-1} + \epsilon_t$$

Variables: c is intercept; ϕ_1 is the AR(1) coefficient.

AR(1) conditional mean

$$\mathbb{E}(y_t \mid F_{t-1}) = c + \phi_1 y_{t-1}$$

Variables: the conditional mean is the one-step forecast given past information.

AR(1) unconditional mean

$$\mathbb{E}(y_t) = \frac{c}{1-\phi_1}$$

Variables: this requires stationarity, typically $|\phi_1| < 1$.

AR(1) conditional variance

$$\mathrm{Var}(\epsilon_t \mid F_{t-1}) = \sigma_\epsilon^2$$

Variables: σ_ϵ^2 is innovation variance.

$$\mathrm{Var}(y_t \mid F_{t-1}) = \sigma_\epsilon^2$$

Variables: uncertainty in y_t conditional on F_{t-1} comes from the new shock.

AR(1) unconditional variance

$$\mathrm{Var}(y_t) = \frac{\sigma_\epsilon^2}{1-\phi_1^2}$$

Variables: this requires stationarity.

AR(1) one-step forecast

$$\mathbb{E}(y_{t+1} \mid F_t) = c + \phi_1 y_t$$

Variables: y_t is known at time t.

AR(1) two-step forecast

$$\mathbb{E}(y_{t+2} \mid F_t) = c + \phi_1 \mathbb{E}(y_{t+1} \mid F_t) = c(1+\phi_1) + \phi_1^2 y_t$$

Variables: the unknown future y_{t+1} is replaced by its forecast.

AR(1) long-horizon forecast

$$\mathbb{E}(y_{t+h} \mid F_t) \to \frac{c}{1-\phi_1}$$

Variables: forecasts mean-revert to the unconditional mean when the AR(1) is stationary.

ARCH and Volatility Models

White noise

$$\mathbb{E}(u_t) = 0$$
$$\mathrm{Var}(u_t) = \sigma^2$$
$$\mathrm{Corr}(u_t, u_{t-k}) = 0, \quad k \geq 1$$

Variables: u_t is a white-noise shock.

MDS conditional mean

$$\mathbb{E}(u_t \mid F_{t-1}) = 0$$

Variables: an MDS is unpredictable from past information.

MDS implies zero unconditional mean

$$\mathbb{E}(u_t) = \mathbb{E}[\mathbb{E}(u_t \mid F_{t-1})] = 0$$

Variables: this uses the law of iterated expectations.

MDS implies no autocovariance

$$\mathrm{Cov}(u_t, u_{t-k}) = 0, \quad k \geq 1$$

Variables: u_{t-k} is known at time $t-1$.

AR(p) model

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

Variables: p is the autoregressive lag order.

AR(p) unconditional mean

$$\mathbb{E}(y_t) = \frac{c}{1-\phi_1-\cdots-\phi_p}$$

Variables: denominator uses the sum of AR coefficients.

AR(p) one-step forecast

$$\mathbb{E}(y_{t+1} \mid F_t) = c + \phi_1 y_t + \phi_2 y_{t-1} + \cdots + \phi_p y_{t-p+1}$$

Variables: all lagged y values on the right are known at time t.

AR(p) selection

$$\text{lowest AIC} = \text{preferred model}$$

Variables: AIC is Akaike information criterion.

Mean equation with residual

$$r_t = \mathbb{E}(r_t \mid F_{t-1}) + \epsilon_t$$

Variables: ϵ_t is the return shock after using information at $t-1$.

Constant-variance benchmark

$$\mathrm{Var}(\epsilon_t \mid F_{t-1}) = \sigma_\epsilon^2$$

Variables: this benchmark has no time-varying conditional volatility.

ARCH LM mean model

$$y_t = x_t'\beta + \epsilon_t$$

Variables: x_t is a regressor vector; β is a coefficient vector.

ARCH LM auxiliary regression

$$\hat{\epsilon}_t^2 = \rho_0 + \rho_1 \hat{\epsilon}_{t-1}^2 + \cdots + \rho_q \hat{\epsilon}_{t-q}^2 + v_t$$

Variables: $\hat{\epsilon}_t^2$ is squared residual; q is the number of ARCH lags tested.

ARCH LM hypotheses and statistic

$$H_0 : \rho_1 = \cdots = \rho_q = 0 \quad \text{no ARCH effects}$$
$$H_1 : \text{at least one } \rho_j \neq 0 \quad \text{ARCH effects exist}$$
$$TR^2 \sim \chi_q^2$$

Variables: R^2 comes from the auxiliary regression.

ARCH LM decision rule

$$TR^2 > \chi_q^2 \text{ critical value} \Rightarrow \text{reject } H_0$$

Variables: rejecting means volatility is time-varying and predictable.

ARCH model structure

$$\epsilon_t = \sigma_t u_t, \qquad u_t \sim \text{iid}(0, 1)$$

Variables: σ_t is conditional standard deviation; u_t is a standardised shock.

ARCH conditional moments

$$\mathbb{E}(\epsilon_t \mid F_{t-1}) = 0, \qquad \text{Var}(\epsilon_t \mid F_{t-1}) = \sigma_t^2$$

Variables: σ_t^2 is conditional variance.

ARCH(1)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2$$

Variables: α_0 is variance intercept; α_1 is the ARCH coefficient.

ARCH(1) conditions

$$\alpha_0 > 0, \qquad \alpha_1 \geq 0, \qquad \alpha_1 < 1$$

Variables: these keep variance positive and finite.

ARCH(q)

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \cdots + \alpha_q \epsilon_{t-q}^2$$

Variables: q is the number of lagged squared shocks.

ARCH(q) conditions

$$\alpha_0 > 0, \qquad \alpha_i \geq 0, \qquad \sum_{i=1}^{q} \alpha_i < 1$$

Variables: $\sum \alpha_i$ measures volatility persistence in ARCH(q).

ARCH(1) unconditional moments

$$\mathbb{E}(\epsilon_t) = 0$$
$$\text{Cov}(\epsilon_t, \epsilon_{t-k}) = 0, \quad k \geq 1$$
$$\text{Var}(\epsilon_t) = \frac{\alpha_0}{1 - \alpha_1}$$

Variables: ARCH errors are uncorrelated, but squared errors can be autocorrelated.

Information criteria

$$AIC = -2\ell + 2k, \qquad BIC = -2\ell + k\log(T)$$

Variables: ℓ is log-likelihood; k is number of estimated parameters; lower is better.